# Calculation Sheet for Structural Load Analysis

## Structure Details

- Structure Type: Simply Supported Reinforced Concrete Beam
- Length (L) = 4 m
- Width (b) = 0.3 m
- Depth (d) = 0.5 m
- Unit Weight of Reinforced Concrete = 25 kN/m$^3$

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## 1. Dead Load Calculation

### Volume of Beam

Volume = Length × Width × Depth

= 4 × 0.3 × 0.5

= 0.6 m$^3$

### Dead Load

Dead Load = Unit Weight × Volume

= 25 × 0.6

= **15 kN**

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## 2. Live Load Calculation

Assumed Live Load = 3 kN/m$^2$

Area = Length × Width

= 4 × 0.3

= 1.2 m$^2$

Live Load = 3 × 1.2

= **3.6 kN**

## 3. Wind Load Calculation

Assumed Wind Load = **1.5 kN**

## 4. Total Load Calculation

Total Load

= Dead Load + Live Load + Wind Load

= 15 + 3.6 + 1.5

= **20.1 kN**

## Summary Table

| Load Type | Value |
|---|---|
| Dead Load | 15 kN |
| Live Load | 3.6 kN |
| Wind Load | 1.5 kN |
| Total Load | 20.1 kN |

### Result

The simply supported reinforced concrete beam is subjected to a **total load of 20.1 kN** and is capable of safely resisting the applied loads under the assumed loading conditions.